

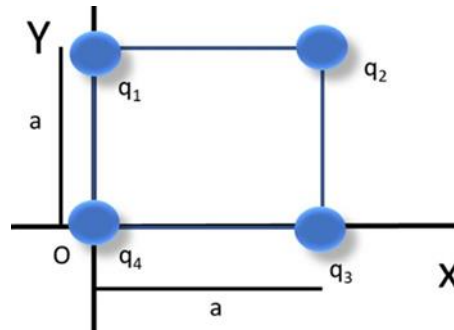
Assignment-2

Course Code: PHY 2105

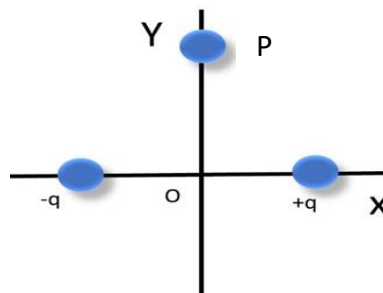
Summer 2026 [262]

Course Title: Physics Content: Coulomb's Law + Electric Field

- Two charges $+16\text{ nC}$ and $+9\text{ nC}$ are placed 50 cm apart. (i) At what point on the line joining them the electric force is zero? (ii) What is the electrostatic force at the midpoint between the two charges if a positive test charge of $+2\text{ nC}$ is introduced?
- Charges of magnitude $10\text{ }\mu\text{C}$ each are located in vacuum at the corners A, B and C of an equilateral triangle measuring 3 meters on each side. If the charge at A and C are negative and the charge B is positive, (i) What is the magnitude and direction of the total force on the charge at B? (ii) Draw the force direction diagram, (iii) What are the x-component and y-component of the resultant forces? and (iv) Express the all forces into vectors.
- What is (i) the Coulomb force between two 4 gm pennies 1 m apart if we remove all the electrons from the $^{23}_{11}\text{Mg}$ atoms? and (ii) What is their acceleration as they separate?
- Three charges lie on the x axis: $q_1=+25\text{ nC}$ at the origin, $q_2=-12\text{ nC}$ at $x=2\text{m}$, $q_3=+18\text{ nC}$ at $x=3\text{ m}$. What is the net force on q_2 ? What is the direction? and Draw the all force directions.
- How many charges are available in (i) 1 kg of electron? and (ii) 1 kg of AlCl_2 ?
- Four charges $q_1 = +2q$, $q_2 = +4q$, $q_3 = +2q$ and $q_4 = -2q$ are placed at the corners of a square. (i) Draw the arrangement of the charges (ii) Calculate the magnitude and direction of electrostatic forces on a charge $+1q$ at the intersection of the diagonals of the square of side 10 cm if $q = 3 \times 10^{-9}\text{ C}$.



- In figure below, two charged particles on an x-axis: $-q = -4\text{ C}$ is at $x = -4\text{ m}$ and $q = 4\text{ C}$ is at $x = 4\text{ m}$. What are the (i) magnitude and direction of the net electric field produced at point P in the Y axis at $y = 4\text{ m}$ and (ii) if $-q = +4\text{ C}$ and at O position $(0,0)$ $+4q$ charge is inserted then also find out the direction and magnitude of the net electric field.



8. An electric dipole with dipole moment $\vec{P} = (4\hat{i} + 5\hat{j} + 7\hat{k})$ Cm is found in an electric field $\vec{E} = (4000\hat{i} + 2000\hat{j} + 3000\hat{k})$ N/C.
- What is the potential energy of the electric dipole?
 - What is the torque acting on it?
 - If an external agent turns the dipole until its electric dipole moment is $\vec{P} = (5\hat{i} - 10\hat{j})$ Cm, how much work is done by the agent?
 - What is the maximum potential energy?
 - What is the work done if dipole moment $\vec{P} = (4\hat{i} + 5\hat{j} + 7\hat{k})$ Cm rotates from 45° to 90° in that system?
9. An electron on the Y axis of an electric dipole is 5 nm from the center of the dipole. What is the magnitude of the electric field and electric force on the electron if the dipole moment is 5.6×10^{-15} Cm? Assume that 5 nm is much larger than the separation of the charged particles that form the dipole.
10. From the figure below, calculate the net electric field required to bring a $+2.2\mu\text{C}$ charge in the center of the shape. The distance of the electrons from the center is 0.1 cm and the distance of proton is 0.6 cm.

